Project: Worked examples
Title: Duct sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Sizes each duct section by the equal-friction method (a target friction rate) or the velocity method (a target velocity), rounds to standard sizes, and reports velocity, friction rate, friction and fitting losses per section and the total pressure drop along the index run, with warnings against user velocity and pressure-drop limits.

WHAT IT CALCULATES	RELATION
Round duct diameter from the design target	D by Colebrook with the air properties at T (equal friction), or D = √(4Q/πv) (velocity), rounded up to the next standard size
Rectangular duct of equal friction and capacity	D_e = 1.30 (ab)^0.625/(a + b)^0.25
Flat-oval duct of equal friction and capacity	D_e = 1.55 A^0.625/P^0.25, with A = $\pi b^2/4$ + b(a − b) and P = πb + 2(a − b)
Velocity method, rectangular or flat-oval duct with side a fixed	b from the free area A = Q/v (rectangle b = A/a; flat oval the smaller root of $(1 - \pi/4)b^2 - ab + A = 0$), rounded up; D_e is kept for the friction only
Friction loss per metre, at the velocity v_e of the equivalent round duct	$\Delta p/L = f\,\rho\,v_e^2/(2D_e)$
Fitting losses, at the actual velocity	$\Delta p = \Sigma C\,\rho\,v^2/2$
Index-run totals, summed over every section of the run	$\Delta p_{run} = \Sigma(\Delta p/L \cdot L + \Sigma C\,\rho\,v^2/2)$

Basis: Duct design (friction chart basis, equivalent diameters, loss coefficients)[1](Ch. 21); Flow of fluids in pipes and ducts[2](Section 4); Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions[3]; Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions[4].

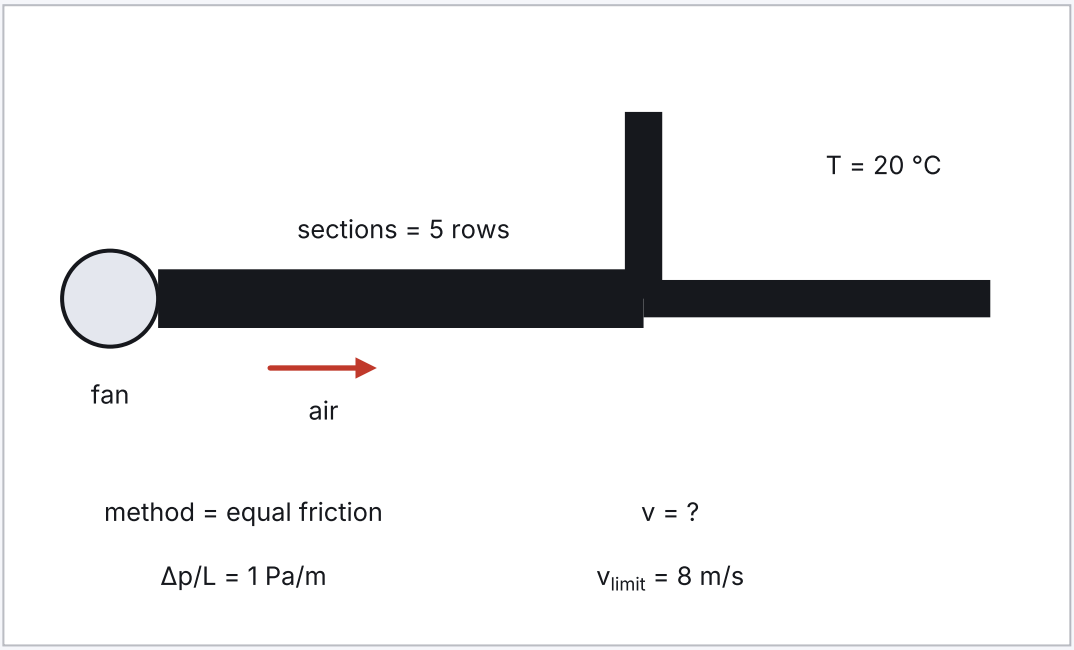


Figure 1 — A duct run from the fan through its sections as listed (a main and two branches drawn), each sized by the chosen method. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Sizing method	equal friction		assumption: Equal friction for a low-pressure system.	any
dp_{target}	Target friction rate (equal friction)	1	Pa/m	spec: SPEC-M-07 §5: 1 Pa/m design friction rate	0.3 … 5 Pa/m
v_{target}	Target velocity (velocity method)	6	m/s	default	2 … 20 m/s
T	Air temperature	20	°C	assumption: Supply air 20 °C.	-20 … 60 °C
p_{atm}	Barometric pressure	101.325	kPa	assumption: Sea level.	80 … 105 kPa
eps	Duct roughness	0.09	mm	spec: Galvanised sheet, medium roughness (ASHRAE Ch. 21 category)	0.03 … 3 mm
sections	Duct sections	5 row(s)		drawing: DRG-M-205 rev B	any
v_{limit}	Velocity limit (warning and check)	8	m/s	spec: SPEC-M-07 §5: 8 m/s in risers	2 … 30 m/s
dp_{limit}	Total pressure-drop limit (check)	250	Pa	spec: Fan external static allowance 250 Pa	50 … 2000 Pa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$round_{to}$	Round sizes up to a multiple of	50	mm	spec: BS EN 1505 / 1506 preferred sizes, 50 mm steps	1 … 100 mm

TABLE ROW	VALUES
sections[1]	name = AHU to riser; Q = 2 m^3/s; L = 15 m; shape = rectangular; a = 800 mm; D = —; C = 0.8; dp_eq = 0 Pa; index = yes
sections[2]	name = riser; Q = 2 m^3/s; L = 12 m; shape = round; a = —; D = —; C = 0.3; dp_eq = 0 Pa; index = yes
sections[3]	name = floor main; Q = 1.2 m^3/s; L = 25 m; shape = rectangular; a = 500 mm; D = —; C = 1.1; dp_eq = 0 Pa; index = yes
sections[4]	name = branch to VAV; Q = 0.4 m^3/s; L = 10 m; shape = round; a = —; D = —; C = 0.9; dp_eq = 60 Pa; index = yes
sections[5]	name = other branch; Q = 0.8 m^3/s; L = 18 m; shape = round; a = —; D = —; C = 1; dp_eq = 60 Pa; index = no

3. Assumptions

A1. Air is dry at the stated temperature and pressure (ρ = p/(287.055 T)); viscosity by Sutherland's law. (Standard air properties.)
A2. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A3. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)
A4. method (Sizing method) is taken as equal friction. (Equal friction for a low-pressure system.)
A5. T (Air temperature) is taken as 20 degC. (Supply air 20 °C.)
A6. p_{atm} (Barometric pressure) is taken as 101.325 kPa. (Sea level.)

4. Working

[10] ρ = p / (287.055 T)
= 101320 Pa / (287.055 × 293.15 K)
= 1.2041 kg/m³
Air density

[11] μ = Sutherland's law
= T = 293.15 K
= 18.133 uPa·s
Air dynamic viscosity

[12] a_{s1} = AHU to riser: side a
= as given
= 800 mm
Side a, AHU to riser

[13] b_{s1} = AHU to riser: side b
= from the target equivalent diameter 573.91 mm
= 400 mm
Side b, AHU to riser

[14] De_{s1} = AHU to riser: equivalent diameter [1](Ch. 21)
= 1.30 (ab)^0.625/(a + b)^0.25
= 609.35 mm
Equivalent diameter, AHU to riser

[15] v_{s1} = AHU to riser: Q / A
= 2 / 0.32
= 6.25 m/s
Velocity, AHU to riser
[16] Re_{s1} = AHU to riser: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.2041 × 6.8581 × 0.60935 / 0.000018133
= 277500
Reynolds number, AHU to riser
[17] f_{s1} = AHU to riser: friction factor
= Colebrook–White
= 0.015999
Friction factor, AHU to riser
[18] dpL_{s1} = AHU to riser: f ρ v_e^2 / (2 D_e)
= 0.015999 × 1.2041 × 6.8581² / (2 × 0.60935)
= 0.74347 Pa/m
Friction rate, AHU to riser
[19] $dp_{f,s1}$ = AHU to riser: (Δp/L) L
= 0.74347 × 15
= 11.152 Pa
Friction loss, AHU to riser
[20] $dp_{fit,s1}$ = AHU to riser: ΣC ρ v²/2
= 0.8 × 23.518
= 18.814 Pa
Fitting loss, AHU to riser
[21] dp_{s1} = AHU to riser: friction + fittings + equipment
= 11.152 + 18.814 + 0
= 29.966 Pa
Section loss, AHU to riser (index run)
[22] D_{s2} = riser: diameter
= target 573.91 mm rounded up to 50 mm
= 600 mm
Diameter, riser
[23] v_{s2} = riser: Q / A
= 2 / 0.28274
= 7.0736 m/s
Velocity, riser
[24] Re_{s2} = riser: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.2041 × 7.0736 × 0.6 / 0.000018133
= 281820
Reynolds number, riser
[25] f_{s2} = riser: friction factor
= Colebrook–White
= 0.015985
Friction factor, riser
[26] dpL_{s2} = riser: f ρ v_e^2 / (2 D_e)
= 0.015985 × 1.2041 × 7.0736² / (2 × 0.6)
= 0.80253 Pa/m
Friction rate, riser
[27] $dp_{f,s2}$ = riser: (Δp/L) L
= 0.80253 × 12
= 9.6303 Pa
Friction loss, riser

[28] $dp_{fit,s2}$ = riser: ΣC ρ v²/2
= 0.3 × 30.124
= 9.0371 Pa
Fitting loss, riser
[29] dp_{s2} = riser: friction + fittings + equipment
= 9.6303 + 9.0371 + 0
= 18.667 Pa
Section loss, riser (index run)
[30] a_{s3} = floor main: side a
= as given
= 500 mm
Side a, floor main
[31] b_{s3} = floor main: side b
= from the target equivalent diameter 473.16 mm
= 400 mm
Side b, floor main
[32] De_{s3} = floor main: equivalent diameter [1](Ch. 21)
= 1.30 (ab)^0.625/(a + b)^0.25
= 488.12 mm
Equivalent diameter, floor main
[33] v_{s3} = floor main: Q / A
= 1.2 / 0.2
= 6 m/s
Velocity, floor main
[34] Re_{s3} = floor main: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.2041 × 6.4127 × 0.48812 / 0.000018133
= 207850
Reynolds number, floor main
[35] f_{s3} = floor main: friction factor
= Colebrook–White
= 0.016905
Friction factor, floor main
[36] dpL_{s3} = floor main: f ρ v_e² / (2 D_e)
= 0.016905 × 1.2041 × 6.4127² / (2 × 0.48812)
= 0.85744 Pa/m
Friction rate, floor main
[37] $dp_{f,s3}$ = floor main: (Δp/L) L
= 0.85744 × 25
= 21.436 Pa
Friction loss, floor main
[38] $dp_{fit,s3}$ = floor main: ΣC ρ v²/2
= 1.1 × 21.674
= 23.841 Pa
Fitting loss, floor main
[39] dp_{s3} = floor main: friction + fittings + equipment
= 21.436 + 23.841 + 0
= 45.277 Pa
Section loss, floor main (index run)
[40] D_{s4} = branch to VAV: diameter
= target 312.82 mm rounded up to 50 mm
= 350 mm
Diameter, branch to VAV

[41] v_{s4} = branch to VAV: Q / A
= 0.4 / 0.096211
= 4.1575 m/s
Velocity, branch to VAV
[42] Re_{s4} = branch to VAV: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.2041 × 4.1575 × 0.35 / 0.000018133
= 96625
Reynolds number, branch to VAV
[43] f_{s4} = branch to VAV: friction factor
= Colebrook–White
= 0.01938
Friction factor, branch to VAV
[44] dpL_{s4} = branch to VAV: f ρ v_e^2 / (2 D_e)
= 0.01938 × 1.2041 × 4.1575² / (2 × 0.35)
= 0.57623 Pa/m
Friction rate, branch to VAV
[45] $dp_{f,s4}$ = branch to VAV: (Δp/L) L
= 0.57623 × 10
= 5.7623 Pa
Friction loss, branch to VAV
[46] $dp_{fit,s4}$ = branch to VAV: ΣC ρ v²/2
= 0.9 × 10.406
= 9.3657 Pa
Fitting loss, branch to VAV
[47] dp_{s4} = branch to VAV: friction + fittings + equipment
= 5.7623 + 9.3657 + 60
= 75.128 Pa
Section loss, branch to VAV (index run)
[48] D_{s5} = other branch: diameter
= target 406.05 mm rounded up to 50 mm
= 450 mm
Diameter, other branch
[49] v_{s5} = other branch: Q / A
= 0.8 / 0.15904
= 5.0301 m/s
Velocity, other branch
[50] Re_{s5} = other branch: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.2041 × 5.0301 × 0.45 / 0.000018133
= 150310
Reynolds number, other branch
[51] f_{s5} = other branch: friction factor
= Colebrook–White
= 0.017809
Friction factor, other branch
[52] dpL_{s5} = other branch: f ρ v_e^2 / (2 D_e)
= 0.017809 × 1.2041 × 5.0301² / (2 × 0.45)
= 0.60286 Pa/m
Friction rate, other branch
[53] $dp_{f,s5}$ = other branch: (Δp/L) L
= 0.60286 × 18
= 10.851 Pa
Friction loss, other branch

[54] $dp_{fit,s5}$ = other branch: ΣC ρ v²/2
= 1 × 15.233
= 15.233 Pa
Fitting loss, other branch
[55] dp_{s5} = other branch: friction + fittings + equipment
= 10.851 + 15.233 + 60
= 86.084 Pa
Section loss, other branch
[56] dp_{index} = Σ section losses on the index run
= 4 section(s)
= 169.04 Pa
Index-run pressure drop
[57] dp_{all} = Σ section losses, all sections
= 5 section(s)
= 255.12 Pa
Sum over every section (not a path)
[58] v_{peak} = highest section velocity
=
= 7.0736 m/s
Highest velocity
[59] u_{dp} = dp_{index} / dp_{limit}
= 169.04 / 250
= 0.67616
Pressure-drop utilisation
[60] u_v = v_{peak} / v_{limit}
= 7.0736 / 8
= 0.88419
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
a_{s1}	800 mm	a, AHU to riser
b_{s1}	400 mm	b, AHU to riser
De_{s1}	609.35 mm	D_e, AHU to riser
v_{s1}	6.25 m/s	v, AHU to riser
Re_{s1}	277500	Re, AHU to riser
f_{s1}	0.015999	f, AHU to riser
dpL_{s1}	0.74347 Pa/m	Δp/L, AHU to riser
$dp_{f,s1}$	11.152 Pa	Friction loss, AHU to riser
$dp_{fit,s1}$	18.814 Pa	Fitting loss, AHU to riser
dp_{s1}	29.966 Pa	Section loss, AHU to riser
D_{s2}	600 mm	D, riser
v_{s2}	7.0736 m/s	v, riser
Re_{s2}	281820	Re, riser
f_{s2}	0.015985	f, riser
dpL_{s2}	0.80253 Pa/m	Δp/L, riser

SYMBOL	VALUE	DESCRIPTION
$dp_{f,s2}$	9.6303 Pa	Friction loss, riser
$dp_{fit,s2}$	9.0371 Pa	Fitting loss, riser
dp_{s2}	18.667 Pa	Section loss, riser
a_{s3}	500 mm	a, floor main
b_{s3}	400 mm	b, floor main
De_{s3}	488.12 mm	D_e, floor main
v_{s3}	6 m/s	v, floor main
Re_{s3}	207850	Re, floor main
f_{s3}	0.016905	f, floor main
dpL_{s3}	0.85744 Pa/m	Δp/L, floor main
$dp_{f,s3}$	21.436 Pa	Friction loss, floor main
$dp_{fit,s3}$	23.841 Pa	Fitting loss, floor main
dp_{s3}	45.277 Pa	Section loss, floor main
D_{s4}	350 mm	D, branch to VAV
v_{s4}	4.1575 m/s	v, branch to VAV
Re_{s4}	96625	Re, branch to VAV
f_{s4}	0.01938	f, branch to VAV
dpL_{s4}	0.57623 Pa/m	Δp/L, branch to VAV
$dp_{f,s4}$	5.7623 Pa	Friction loss, branch to VAV
$dp_{fit,s4}$	9.3657 Pa	Fitting loss, branch to VAV
dp_{s4}	75.128 Pa	Section loss, branch to VAV
D_{s5}	450 mm	D, other branch
v_{s5}	5.0301 m/s	v, other branch
Re_{s5}	150310	Re, other branch
f_{s5}	0.017809	f, other branch
dpL_{s5}	0.60286 Pa/m	Δp/L, other branch
$dp_{f,s5}$	10.851 Pa	Friction loss, other branch
$dp_{fit,s5}$	15.233 Pa	Fitting loss, other branch
dp_{s5}	86.084 Pa	Section loss, other branch
dp_{index}	169.04 Pa	Index-run Δp
dp_{all}	255.12 Pa	Σ all sections
v_{peak}	7.0736 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Velocity ≤ limit	u_v = 0.88419	u_v ≤ 1	0.884	
PASS	Index-run Δp ≤ limit	u_{dp} = 0.67616	u_{dp} ≤ 1	0.676	

7. Applicability

WARNING — input.v_target: v_{target} (Target velocity (velocity method)) was not entered; the template default 6 m/s is used.

8. Verification

```
Template bs.duct-sizing@1.1.0 Content hash 82e9365c81cc4726 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
3. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Rectangular Cross-Section—Dimensions. BS EN 1505. BSI.
4. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Circular Cross-Section—Dimensions. BS EN 1506. BSI.